SpaceX Falcon 9
First stage Landing Prediction

Applied Machine Learning Pipeline to Forecast Booster Reusability & Launch Economics

Capstone Project Report | Data Science & Predictive Analytics

Outline

Executive Summary

Core Objective

Develop an end-to-end classification pipeline to predict whether the SpaceX Falcon 9 first stage booster will successfully land, enabling competitive cost benchmarking against traditional launch providers.

Key Outcomes

Achieved 83.3% test classification accuracy across evaluated models. Identified key success drivers including payload mass thresholds, orbit dynamics, and launch site geographic parameters.

Introduction

$62M

SpaceX Commercial Launch Price

$165M

Traditional launch providers

The Reusability Advantage

Traditional launch providers incur costs upward of **$165 million** per launch because rockets are single-use.

SpaceX delivers over **60% cost savings** by recovering and refurbishing the Falcon 9 first stage booster, revolutionizing commercial space transportation market dynamics.

Data Science Methodology

2. Wrangling

Data normalization, missing value imputation, & binary target class creation.

4. Prediction models

Hyperparameter tuning for Logistic Regression, SVM, Decision Tree & KNN.

1. Collection

SpaceX REST API querying & Wikipedia web scraping with BeautifulSoup.

3. EDA & Maps

SQL query analysis, Seaborn visualizations, Folium maps & Dash dashboard.

Data Collection Methodology

SpaceX REST API

Extracted raw launch details, booster versions, payload specifications, launchpads, and landing outcomes across historical missions.

Pulling SpaceX Launch Data from Wikipedia

- **Grab Falcon 9 and Falcon Heavy launch history** straight from Wikipedia.
- **Pull up the web page** using the link for Falcon 9's launch records.
- **Snag the column headers** from the table to see what data fields we're working with.
- **Turn the table data into a clean DataFrame** so it’s ready to analyze.

Data Wrangling Methodology

Web Scraping

Scraped supplementary Falcon 9 historical records from Wikipedia using BeautifulSoup to validate landing pad coordinates and mission payloads.

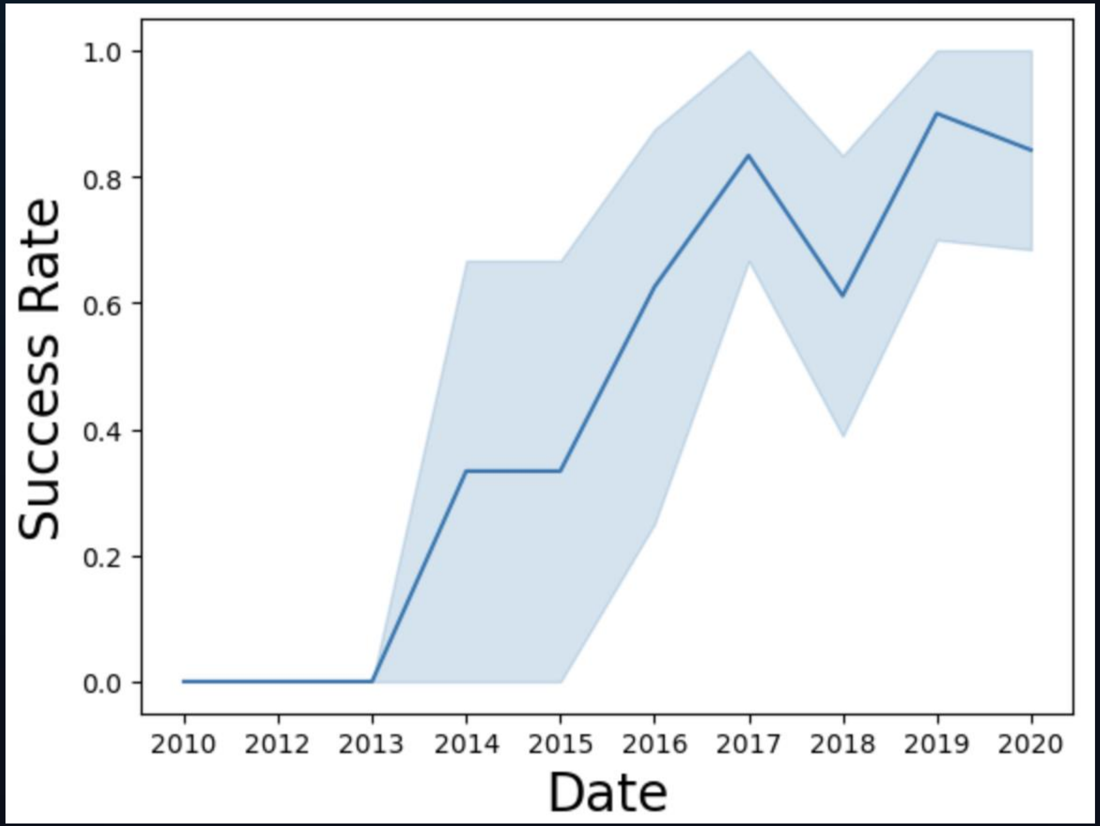
Cleaning & Preparing the Launch Data

- **Count up launches by location** to see which sites are used the most.
- **Break down orbit types** to see how frequently each target orbit is targeted.
- **Track success rates per orbit** to check how mission outcomes vary depending on the destination.
- **Build a landing success label** using raw data from the main Outcome column.

Predictive Analysis Methodology

- **Extract the target variable:** Grab the 'Class' column as a Pandas Series, convert it to a NumPy array, and store it in Y.
- **Scale the feature set:** Standardize the feature matrix X so all variables are on the same scale, then overwrite X.
- **Split the data:** Divide features and labels into training and testing sets (holding back 20% for testing with `random_state=2` for consistency).
- **Tune Logistic Regression:** Set up Logistic Regression with 10-fold cross-validation (`GridSearchCV`) to test hyperparameters and find the best-performing configuration.

EDA with Visualization



Visualised launch success rate throughout the years

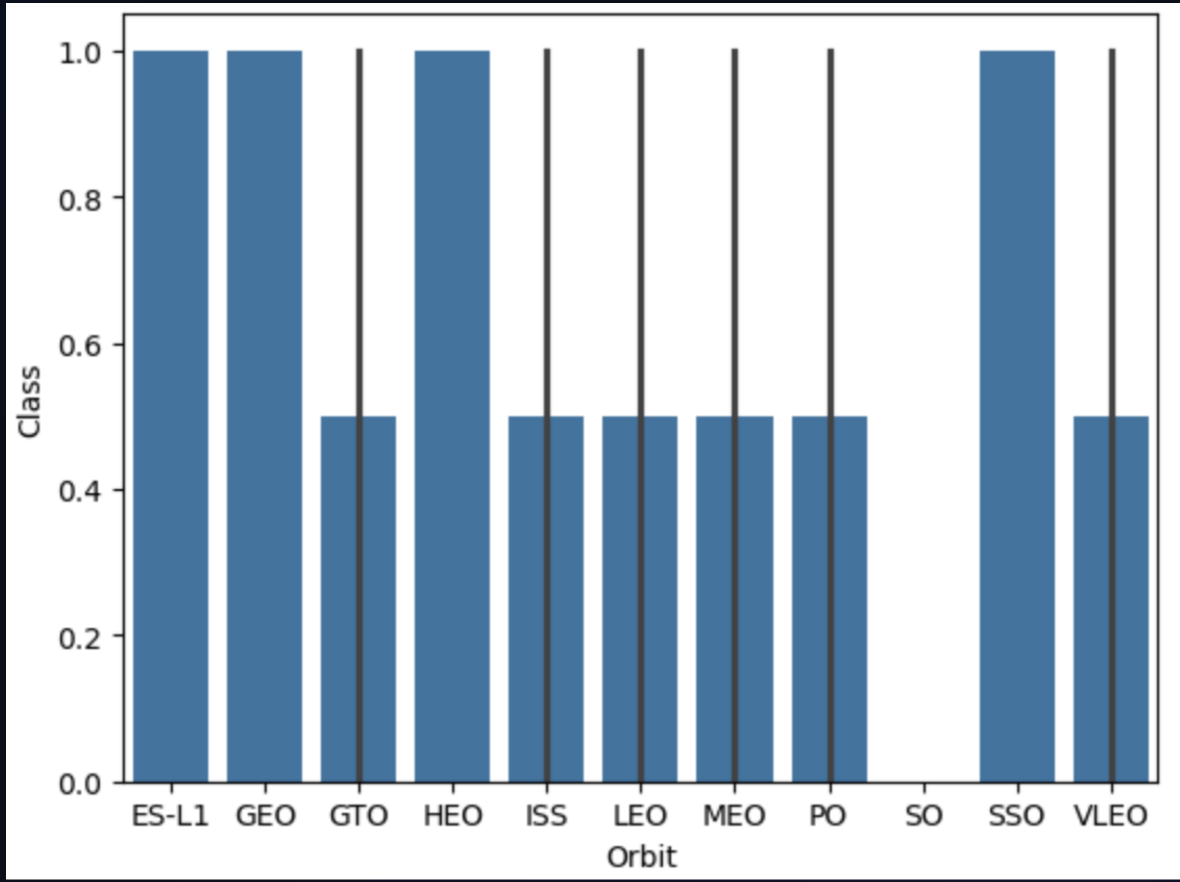
EDA with Visualization

Goals

- Plot the success rate associated with each target orbit type.
- Contrast launch outcomes across various orbit categories.

Key Findings

- Mission success varies noticeably depending on the intended orbit type.
- Low Earth Orbit (LEO) and ISS-bound runs achieved particularly strong success rates.
- The bar chart provides a direct comparison of success performance across every orbit destination.



Data Wrangling with SQL

Exploring the Data with SQL Queries

- **Find all unique launch pads** used across the missions.
- **Pull 5 sample records** for launch sites starting with "CCA".
- **Calculate total payload weight** delivered for NASA Commercial Resupply Services (CRS) missions.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

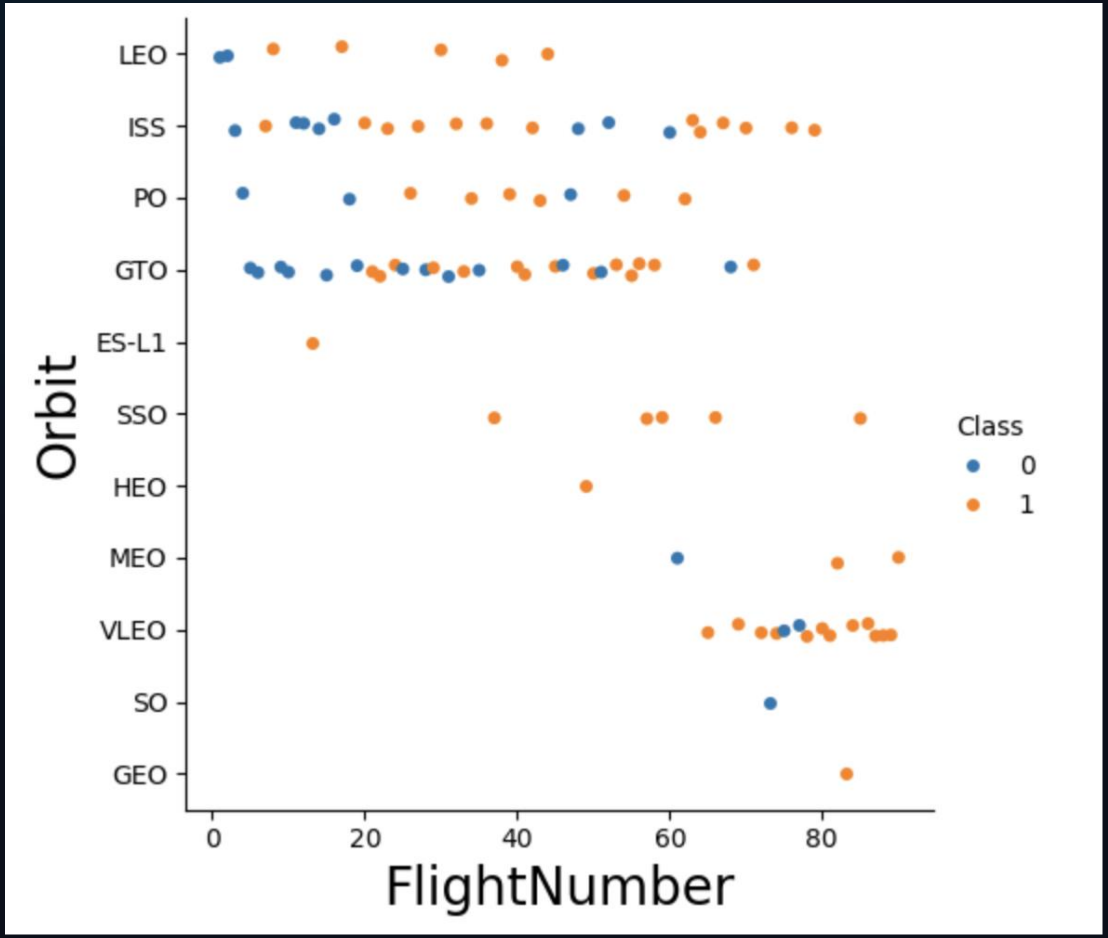
Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- **Calculate average payload weight** specifically for the F9 v1.1 booster.
- **Find the date** of the very first successful ground-pad landing.
- **Identify boosters that landed on a drone ship** while carrying a payload between 4,000 kg and 6,000 kg.
- **Count up the overall wins and losses** by totaling successful vs. failed mission outcomes.

- **Find the heaviest payloads carried** and list the booster versions responsible using a subquery.
- **Filter for 2015 drone ship failures**, pulling up the month, booster version, and launch site for each instance.
- **Rank landing successes by frequency** between June 4, 2010, and March 20, 2017 (highest to lowest).

avg(PAYLOAD_MASS__KG_)
2928.4

Flight number vs Orbit type



Goals

- Map out the connection between Flight Number and target Orbit Type visually.
- Contrast how flight attempts are spread across various orbit categories.

Key Findings

- Flight numbers steadily rise as SpaceX executes consecutive missions over time.
- Launches targeted a wide variety of orbits, including LEO, ISS, GTO, and Polar paths.
- The scatter plot clearly shows how flight counts are distributed across each orbit classification.

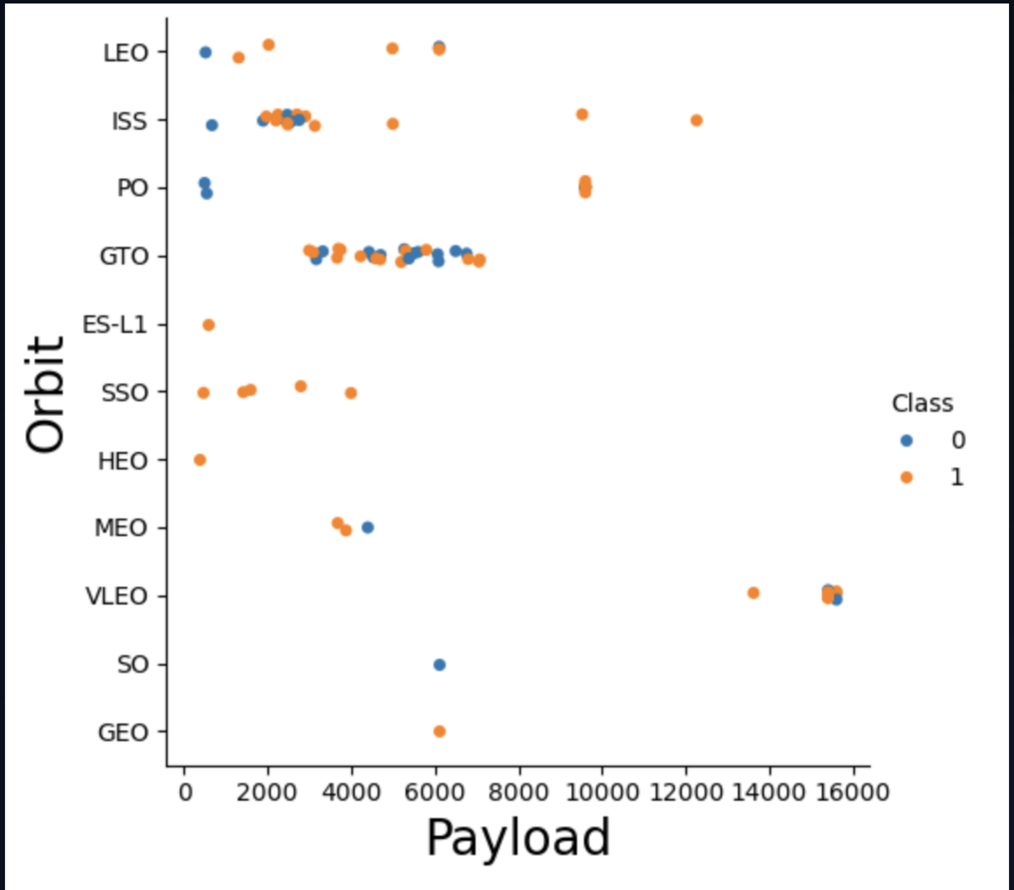
Payload vs Orbit type

Goals

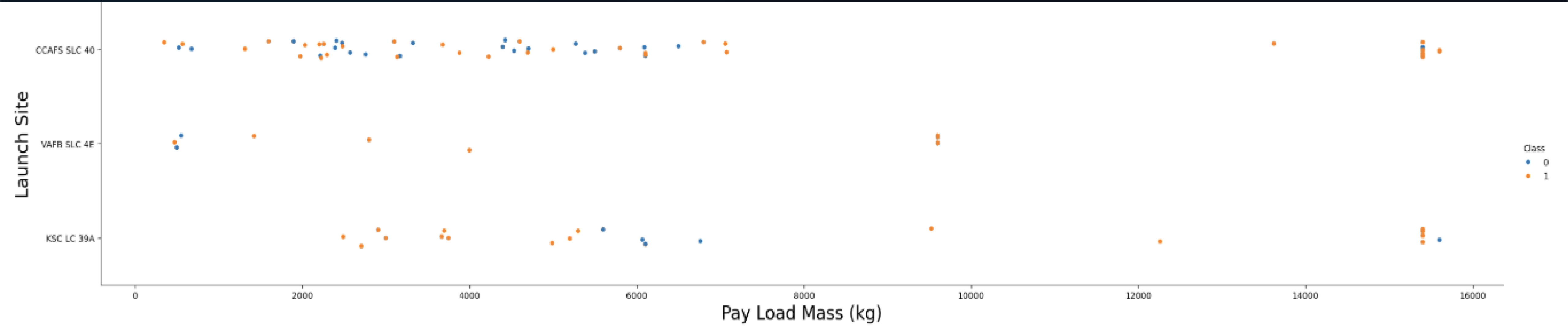
- Graph the correlation between Payload Mass and target Orbit Type.
- Contrast how payload weight is distributed across various orbit categories.

Key Findings

- Payload weights varied significantly depending on the target orbit type.
- Certain orbit destinations accommodated both light and heavy payload capacities.
- The scatter plot clearly reveals payload weight trends across different orbit classifications.

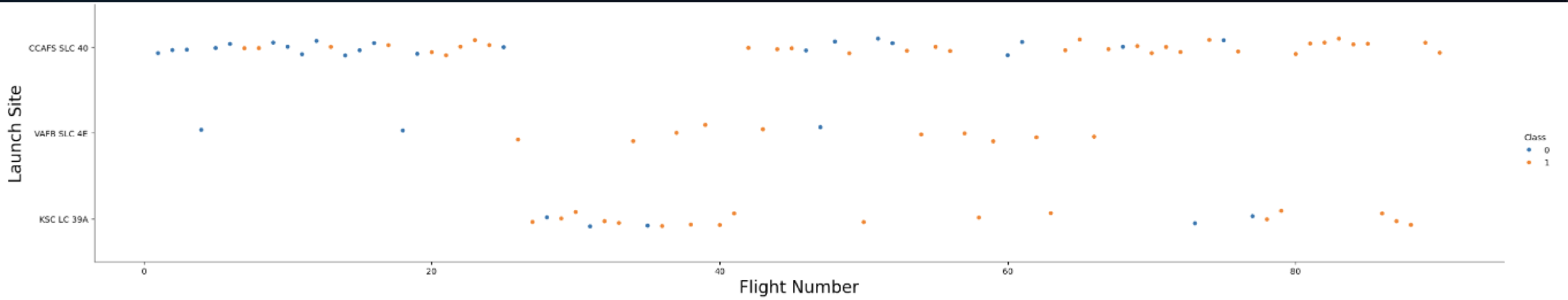


Payload vs Launch site



If you look at the scatter plot comparing payload mass to launch sites, you'll notice that VAFB-SLC hasn't launched any heavy payloads over 10,000 kg.

Flight number vs Launch site



Success rates consistently climbed across all three launch sites as flight numbers increased. For example, VAFB SLC-4E hit a perfect 100% success rate past Flight 50, whereas KSC LC-39A and CCAFS SLC-40 achieved 100% success after their 80th flight.

First Successful Ground Landing Date

Goals

- Pinpoint the exact date of SpaceX's initial successful ground pad landing.
- Identify when the company achieved its first successful booster touchdown on land.

min(DATE)
2015-12-22

Key Findings

- The maiden successful ground pad touchdown occurred on December 22, 2015.
- This event represented a major breakthrough for reusable Falcon 9 first-stage boosters.

First Successful Ground Landing Date

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Goals

- Compute the overall count of successful versus failed mission results.
- Contrast how frequently each specific mission outcome occurs.

Key Findings

- The database query aggregates total totals for every launch outcome.
- The vast majority of flights recorded in the dataset ended in success.

Data Wrangling with SQL

Goals

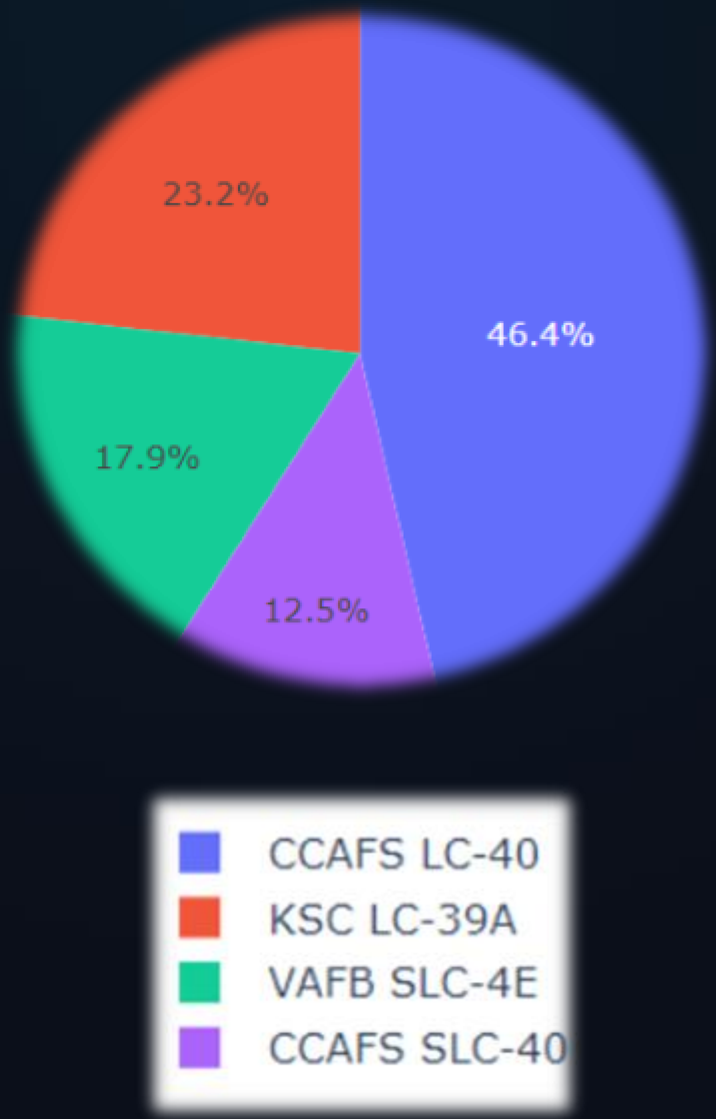
- Pinpoint which booster versions launched the heaviest overall payload masses.
- Compile a complete list of every booster in the dataset operating at peak payload capacity.

Key Findings

- The query highlighted 12 specific booster versions that carried maximum payload weights.
- Each of these top-tier boosters lifted the maximum recorded payload mass of 15,600 kg.
- A majority of these high-capacity boosters were part of the Falcon 9 Block 5 (F9 B5) fleet.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Launch Outcome Proportions by all Sites with Dash



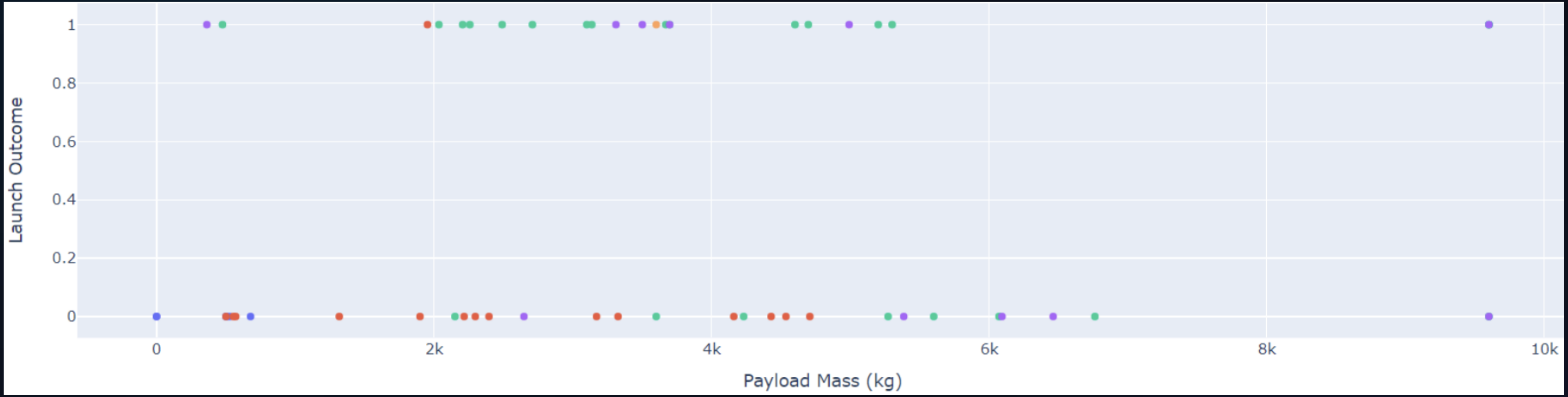
Core Elements

- The dashboard includes a pie chart mapping out success totals for every site.
- Each chart segment represents an individual launch location.
- The area of each slice reflects the total volume of successful launches from that specific pad.

Key Observations

- The chart offers a direct breakdown of successful flights across all locations.
- KSC LC-39A and CCAFS SLC-40 account for the largest share of successful missions.
- VAFB SLC-4E records a noticeably lower portion of successful launches.

Correlation between payload mass and Launch Success with Dash



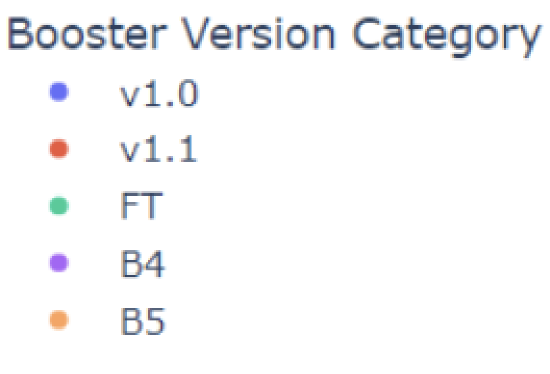
Findings

• Successful launches (Class = 1) occur across multiple payload ranges.

• Different booster versions show varying success patterns.

• The payload slider allows filtering and comparison of launch outcomes.

• KSC LC-39A Contributes the largest share of successful launches in the dashboard.



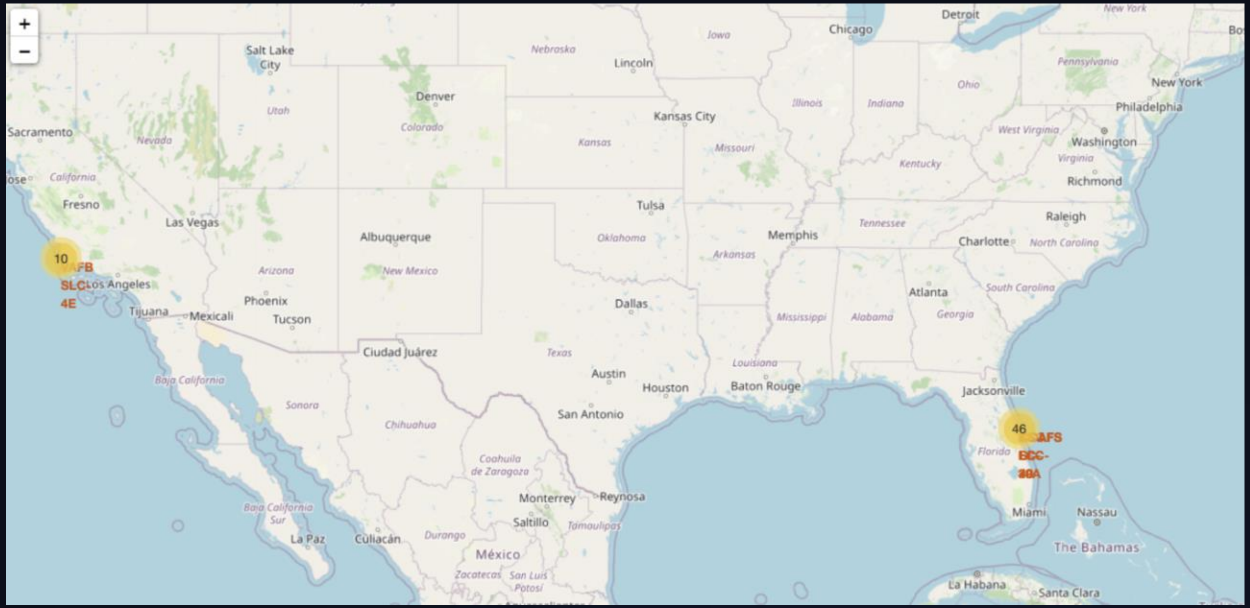
Interactive Visual Analytics with Folium

Goal

- Render every SpaceX launch site on a world map to clearly illustrate where facilities are geographically situated.
- Build a clear visual representation of how launch pads are distributed using the Folium library.

Key Highlights

- The map features interactive markers that pinpoint the precise location of every SpaceX site.
- Every individual pin highlights a specific launch facility.

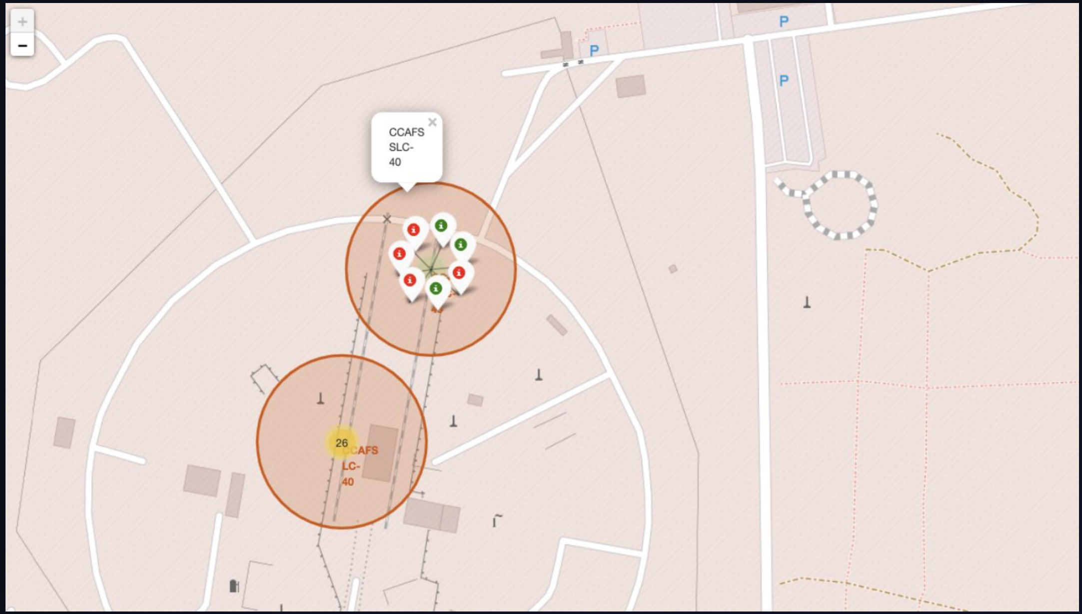


- **US-Based Operations:** The vast majority of launch facilities are positioned within the United States.
- **East Coast Hub:** Florida hosts three major sites—CCAFS LC-40, CCAFS SLC-40, and KSC LC-39A.
- **West Coast Facility:** A single primary facility, VAFB SLC-4E, operates out of California.

Interactive Visual Analytics with Folium

Objectives
- Represent mission results using distinct color-coded map pins.
- Map out the wins and failures associated with each specific launch pad.

Core Features
- Green pins highlight missions that landed successfully.
- Red pins mark launches that ended in failure.
- Cluster groups bundle nearby launch entries together to keep the map readable.



•**High Win Rate:** Nearly every launch site accounts for more successful missions than failures.
•**Positive Trend:** Green markers easily outnumber red ones, pointing to a strong overall launch track record.
•**Cleaner Visuals:** Grouping pins into clusters cleans up the map by stopping overlapping entries from cluttering the view.

Predictive Models Evaluation

Classification Algorithm	Best Hyperparameters	Train Accuracy	Test Accuracy
Decision Tree Classifier	criterion: entropy, max_depth: 4, splitter: random	84.7%	**83.3%**
Logistic Regression	C: 0.01, penalty: l2, solver: lbfgs	84.7%	**83.3%**
Support Vector Machine (SVM)	C: 1.0, kernel: rbf, gamma: scale	84.7%	**83.3%**
K-Nearest Neighbors (KNN)	n_neighbors: 5, algorithm: auto, p: 2	84.7%	**83.3%**

All optimized classifiers achieved an identical top test accuracy of 83.33%, demonstrating reliable generalization on unseen test sets.

Key Machine Learning Insights

- Payload Mass Impact: Light payloads (< 5,300 kg) exhibit significantly higher landing success rates across all launch sites.
- Flight Count Progression: Booster reliability improved drastically after Flight #20 as block revisions (Block 5) matured.
- Coastal Proximity: All successful drone ship and land pad recovery sites are strategically placed along coastlines for trajectory safety.
- Model Generalization: Hyperparameter tuning via GridSearchCV successfully prevented overfitting across all classification algorithms.

Strategic Value & Market Impact

Competitive Bidding

By predicting booster landing probability for target missions, rival launch startups can accurately model SpaceX launch margins and formulate competitive commercial bids.

Reusability Forecasting

Data-driven predictive pipelines offer essential risk management insights for satellite operators deciding between re-flown or new booster stages.

Questions & Discussion

SpaceX Falcon 9 First Stage Landing Prediction Project

IBM Data Science Capstone Presentation | Machine Learning & Predictive Modeling

Conclusion

•**Launch Site Distribution:** Operations are concentrated in the US, driven mainly by Florida's East Coast hubs alongside California's West Coast facility.

•**Launch Success Rates:** Successful missions heavily outnumber failures overall, led by top-performing pads KSC LC-39A and CCAFS SLC-40.

•**Orbit & Payload Analysis:** Operational reliability peaked in LEO and ISS targets, with payload capacities dynamically scaling across orbit types.

•**Milestones & Capability:** Landmark achievements—like the first 2015 land landing—and 15,600 kg max-payload flights confirm the reliability of the reusable Falcon 9 Block 5 fleet.

•**Predictive Modeling:** All evaluated classification models achieved an identical **83.33% accuracy score** in predicting landing success on test data.

Appendix

- Github Repository: https://github.com/dimi-kaffes/Data-analysis-capstone

- https://www.imaging-resource.com/news/how-the-most-prolific-modern-launch-photographer-catches-rockets/